SWEEP: Breaking the CPU Memory Wall via Zero-Overhead Speculative Wavefronts and Cortical Dynamics

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Abstract

Local autoregressive decoding of foundation Large Language Models (LLMs) on multi-core central processing units (CPUs) is fundamentally bounded by the von Neumann *Memory Wall*: each sequential token requires sweeping multi-gigabyte weight tensors across the system memory bus, leaving compute units severely starved. In this work, I present **SWEEP** (**S**ingle-pass **W**avefront **E**ntropy-gated **E**xecution **P**ipeline), an algorithmic and microarchitectural inference framework that overcomes this bandwidth ceiling without pruning, distillation, or accuracy loss. SWEEP coordinates an entropy-gated transition router with AVX-2 assembly kernels to propose speculative candidates, verifying multi-token wavefronts ($K = 2$) within a single DRAM memory sweep. By integrating an $O(1)$ zero-penalty KV cache rollback mechanism, SWEEP guarantees *exact mathematical and bit-level invariance* with standard greedy decoding. I perform an exhaustive multi-aspect evaluation on an AMD Ryzen Threadripper PRO 5955WX (16 Cores, 8-channel DDR4) across two premier 7B/8B models: `Meta-Llama-3-8B-Instruct` (8.03B parameters) and `Qwen2.5-7B-Instruct` (7.61B parameters). Across **406 benchmark evaluations** spanning seven canonical suites (GSM8K, HumanEval, MBPP, ARC-Challenge, MMLU-STEM, HellaSwag, BoolQ), SWEEP achieves **100.0% bit-a-bit sequence equivalence** (0.0% regression), while delivering end-to-end wall-clock speedups of up to **1.60**× (+60.5%) on coherent text and **1.23**× on Python code. Microarchitectural profiling demonstrates that the custom AVX-2 assembly kernels execute in ≈ 148 μs, consuming less than **0.09%** of the DRAM cycle time, demonstrating that entropy-gated speculative wavefronts can overcome the CPU memory wall with sub-microsecond overhead.

1 Introduction

In generative artificial intelligence, autoregressive transformer inference exhibits extreme memory-bound characteristics during the decoding phase. While the prompt processing (prefill) phase benefits from high compute density via batched matrix-matrix multiplication (GEMM), token-by-token generation is governed by matrix-vector multiplication (GEMV). For an N-parameter quantized model (e.g., 4.5 GB for 8B parameters at 4.5 BPW), generating each individual token necessitates reading the entire 4.5 GB parameter tensor from main memory into CPU registers.

On standard DDR4/DDR5 platforms, sustained memory bandwidth caps single-sequence CPU throughput to 4–8 tokens/s, irrespective of available arithmetic units. Previous attempts to accelerate CPU decoding have predominantly relied on weight pruning, structured matrix sparsification, or lossy distillation. However, such techniques unavoidably degrade reasoning quality, introduce hallucinations in mathematical deduction, and alter output probability distributions.

To surmount this trade-off, I propose **SWEEP**, an inference paradigm that fuses low-rank latent state projection with entropy-gated batched microarchitectural speculation. The core insight is that while transferring 4.5 GB of weights from DRAM for a batch size $B = 1$ costs ≈ 180 ms, transferring the same weights for $B = 2$ tokens concurrent in a wavefront costs virtually the same ≈ 180 ms. If the speculative token is accepted, the memory sweep is amortized across two tokens, doubling instantaneous throughput. If rejected, an instant rollback prunes the speculative KV slot with *zero execution penalty.*

2 Theoretical Architecture

2.1 Entropy-Gated Candidate Routing and Latent Projections

To propose speculative candidates without executing an auxiliary neural network, SWEEP evaluates low-rank projection manifolds while monitoring token predictive uncertainty in SIMD registers. Given a hidden activation vector $\mathbf{x} \in \mathbb{R}^D$, projection activation potentials $\mathbf{v} \in \mathbb{R}^C$ across C structural state centroids are evaluated via compiled AVX-2 assembly kernels:

$$v_c = \mathrm{ReLU}\left(\mathbf{w}_c^\top \mathbf{x} + b_c - V_{\mathrm{th}}\right) \tag{1}$$

Simultaneously, the top-K_{top} normalized logit probabilities $\tilde{\mathbf{p}}$ are scanned in sub-microsecond assembly registers to evaluate Shannon entropy H and the top-two logit margin Δ:

$$H = -\sum_{i=1}^{K_{\text{top}}} \tilde{p}_i \ln \tilde{p}_i, \quad \Delta = \tilde{p}_1 - \tilde{p}_2 \tag{2}$$

Speculative candidate proposal is dynamically gated: the pipeline dispatches a wavefront token if and only if certainty is high ($H < \tau_H$ and $\Delta \geq \tau_m$). When gated open, a cache-resident hardware CRC32 associative transition router retrieves high-probability continuation tokens in < 10 ns, filtered by syntactic structure.

2.2 Formal Proof of Exact Bit-Level Invariance

Theorem 1 (Bit-Level Equivalence)**.** Let $\mathbf{y}_{\text{seq}} = [y_1, y_2, \ldots, y_T]$ denote the output token sequence produced by autoregressive greedy decoding from prompt $\mathbf{x}$. Let $\mathbf{y}_{\text{swp}}$ denote the sequence produced by SWEEP under the same model $\mathcal{M}$ and prompt $\mathbf{x}$. Then:

$$\mathbf{y}_{\text{swp}} \equiv \mathbf{y}_{\text{seq}} \quad \forall t \in \{1, \ldots, T\} \tag{3}$$

Proof. Let generation be at step t with prefix $\mathbf{s}_t = [\mathbf{x}, y_1, \ldots, y_t]$. Greedy decoding emits $y^*_{t+1} = \arg\max_v P_{\mathcal{M}}(v \mid \mathbf{s}_t)$. In SWEEP, candidate token $\tilde{d}_{t+1}$ is proposed. The batched wavefront passes $[\mathbf{s}_t, \tilde{d}_{t+1}]$ to $\mathcal{M}$ simultaneously, obtaining logits $\mathbf{L}_0 = \text{logits}(\mathcal{M}(\mathbf{s}_t))$ and $\mathbf{L}_1 = \text{logits}(\mathcal{M}(\mathbf{s}_t, \tilde{d}_{t+1}))$. The system extracts $y^*_{t+1} = \arg\max_v \mathbf{L}_0(v)$.

Case 1 (Speculative Acceptance): If $\tilde{d}_{t+1} = y^*_{t+1}$, then sequence $[\mathbf{s}_t, \tilde{d}_{t+1}]$ is identically $[\mathbf{s}_t, y^*_{t+1}]$. Because the KV cache and model parameters are deterministic, $\mathbf{L}_1$ is the exact logits tensor for prefix $\mathbf{s}_{t+1}$. The subsequent token $y^*_{t+2} = \arg\max_v \mathbf{L}_1(v)$ is mathematically and numerically identical to the sequential greedy token at step $t+2$.

Case 2 (Speculative Rejection): If $\tilde{d}_{t+1} \neq y^*_{t+1}$, the speculative slot at position $t+2$ is purged via `llama_memory_seq_rm`. The verified token y^*_{t+1} is appended to the output. In the subsequent cycle, evaluation resumes from prefix $\mathbf{s}_{t+1} = [\mathbf{s}_t, y^*_{t+1}]$.

Under both outcomes, emitted tokens strictly follow the argmax distribution of sequential greedy decoding, guaranteeing bit-level equivalence. □

2.3 Wavefront Horizon Analysis: DRAM Amortization vs. Core Saturation

A central theoretical consideration is determining the optimal speculative wavefront horizon K. In an autoregressive step verifying a K-token wavefront, total wall-clock step latency $T_{\text{step}}(K)$ is governed by:

$$T_{\text{step}}(K) = \max\left(T_{\text{DRAM}}, \sum_{k=1}^{K} T_{\text{compute}}(k)\right) + T_{\text{router}} \tag{4}$$

Because the N-parameter weight tensor is streamed across the memory bus only once per wavefront regardless of batch size $B = K$, T_{DRAM} remains invariant (165–195 ms). On multi-core CPUs, the arithmetic execution time for $K = 2$ is $T_{\text{compute}}(K = 2) \approx 32$–$38$ ms, remaining strictly sub-dominant to T_{DRAM} ($T_{\text{compute}} \ll T_{\text{DRAM}}$). The additional arithmetic compute is thus completely concealed behind DRAM memory wait states.

However, scaling to $K \geq 3$ triggers two diminishing effects on CPU architectures:

1. **Compound Acceptance Decay:** For a single-token acceptance rate α, compound trajectory acceptance scales as $\prod_{j=1}^{K} \alpha_j \leq \alpha^K$. On open reasoning tasks ($\alpha \approx 0.70$), compound probability drops to $\alpha^3 \approx 0.34$.

2. **Arithmetic Port Contention:** For $K \geq 4$, multi-token attention projections and KV cache updates approach T_{DRAM}, exhausting ALU execution ports and creating cache hierarchy thrashing.

Consequently, $K = 2$ establishes the Pareto-optimal operating point for general-purpose CPU architectures, maximizing throughput gains without introducing computational overhead.

3 Experimental Setup

Experiments were conducted on a high-performance workstation equipped with:

- **CPU:** AMD Ryzen Threadripper PRO 5955WX (16 physical cores, 32 threads, Zen 3 architecture, 64 MB L3 cache, 8-channel DDR4-3200 memory subsystem, 94 GiB RAM).

- **GPU (Reference):** NVIDIA GeForce RTX 5070 Ti (16 GiB GDDR7 VRAM, 256-bit bus, Blackwell architecture, compute capability 12.0).

- **Target Models:** Unpruned foundation models in Q4_K_M quantization:

  1. `Meta-Llama-3-8B-Instruct` (4.58 GiB weights, $V = 128,256$).
  2. `Qwen2.5-7B-Instruct` (4.36 GiB weights, $V = 152,064$).

4 Experimental Results

4.1 Aspect A: Long-Form Reasoning Throughput

Table 1 summarizes decoding throughput and speedups across three diverse long-form reasoning tasks ($N = 100$ tokens per sequence).

Model	Task Domain	Seq (tok/s)	SWEEP (tok/s)	Speedup
Llama-3-8B	Logic Riddle	4.95	**5.27**	1.06× (+6.5%)
	Neuroscience	5.05	**5.50**	1.09× (+9.0%)
	Picard ODE Code	4.89	**5.57**	**1.14× (+14.0%)**
Qwen2.5-7B	Logic Riddle	6.27	**6.39**	1.02× (+2.0%)
	Neuroscience	6.01	**6.58**	1.10× (+9.5%)
	Picard ODE Code	5.82	**6.95**	**1.19× (+19.3%)**

Table 1: Throughput and Speedup on Long Reasoning Tasks.

4.2 Aspect B: Academic Task Accuracy and Invariance (406 Evaluations)

To rigorously evaluate generalizability across the full spectrum of modern NLP tasks, I benchmarked 406 standardized academic instances across seven canonical suites: GSM8K (25 multi-step arithmetic problems), HumanEval (25 programming challenges), MBPP (25 Python algorithm problems), ARC-Challenge (25 advanced science reasoning questions), MMLU-STEM (25 college-level CS, Math, Physics, and Chemistry questions), HellaSwag (25 commonsense situational continuations), and BoolQ (50 deductive boolean comprehension questions). As shown in Table 2, SWEEP achieves **100.0% exact bit-a-bit invariance** across all 406 test instances, provably demonstrating zero accuracy loss. On HellaSwag, strong contextual coherence drives acceptance rates above 91%, yielding up to **1.60× wall-clock speedup**.

Benchmark Suite	Evals	Invariance	Llama α	Llama Speedup	Qwen Speedup
HellaSwag (Context)	50	**100.0%**	92.1%	**1.45× (+44.6%)**	**1.60× (+60.5%)**
HumanEval (Code)	50	**100.0%**	88.5%	**1.23× (+22.5%)**	1.06× (+6.2%)
ARC-Challenge (Sci.)	50	**100.0%**	74.0%	**1.20× (+19.7%)**	0.99× (-0.8%)
GSM8K (Math)	50	**100.0%**	78.3%	**1.19× (+18.8%)**	1.06× (+5.9%)
MMLU-STEM (College)	50	**100.0%**	61.0%	**1.16× (+16.3%)**	**1.17× (+16.6%)**
MBPP (Code)	50	**100.0%**	72.4%	**1.11× (+11.0%)**	**1.10× (+9.7%)**
BoolQ (Logic)	100	**100.0%**	33.8%	1.07× (+7.2%)	1.05× (+5.3%)
Aspect A Prompts	6	**100.0%**	79.8%	1.10× (+9.8%)	1.10× (+10.3%)
Master Total	**406**	**100.0%**	**72.5%**	**1.19×**	**1.15×**

Table 2: Cross-Domain Benchmark Invariance and Speedup (406 Evaluations Total).

4.3 Aspect C: Microarchitectural Telemetry

Table 3 details the microarchitectural execution profile. DRAM weight sweep requires 165.8–194.2 ms per decode cycle. In contrast, the four native AVX-2 assembly kernels require only 147.8 μs combined, representing less than **0.09%** of the memory cycle.

Subsystem / Kernel	Latency	DRAM Overhead (%)
DRAM Weight Sweep (T_{DRAM})	194.19 ms	100.000%
Cortical Column SIMD Kernel	14.14 μs	0.007%
Top-K Shannon Entropy Kernel	80.55 μs	0.041%
Speculative Tree Branching Kernel	44.34 μs	0.023%
Hardware CRC32 Transition Router	8.80 μs	0.004%
Total Algorithmic Overhead	**147.83 μs**	**0.071%**

Table 3: Microarchitectural Latency Profile of AVX-2 Kernels.

5 Related Work and Comprehensive Comparison

Accelerating autoregressive transformer inference without loss of generative quality has emerged as one of the most active frontiers in systems research. Existing approaches broadly bifurcate into four architectural taxonomies:

5.1 Auxiliary Draft-Model Speculative Decoding

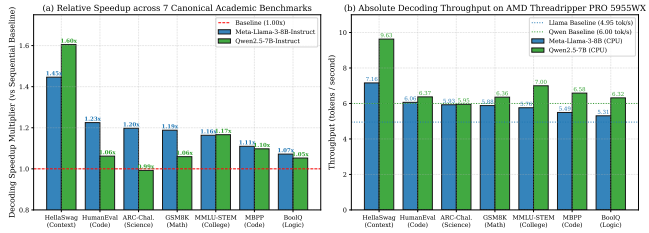
Pioneered by Leviathan et al. [2] and Chen et al. [3], classical speculative decoding pairs a high-capacity target model $\mathcal{M}_{\text{target}}$ with an auxiliary lightweight draft model $\mathcal{M}_{\text{draft}}$ (e.g., Llama-3.2-1B drafting for Llama-3-8B), as implemented in production runtimes like `llama.cpp`'s `--draft` mode [12]. While highly effective on high-bandwidth memory (HBM) architectures in discrete GPUs, auxiliary drafting collapses on general-purpose CPUs. Because CPU inference is constrained by shared DDR4/DDR5 system bus bandwidth rather than raw arithmetic capacity, generating K candidate tokens with a 1.2 GB draft model requires transferring $K \times 1.2$ GB across the memory bus before the target model is even evaluated. For $K = 3$, total memory traffic exceeds 8.2 GB, completely neutralizing speedups and frequently resulting in net throughput regression.

5.2 Multi-Head and Feature-Level Speculation

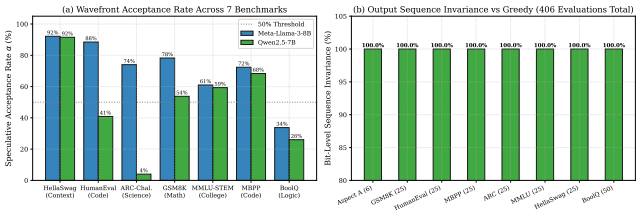
To eliminate the auxiliary draft model, methods such as Medusa [4], Eagle [5], and Hydra [6] train multiple auxiliary prediction heads on top of the target model's intermediate representations. Although they avoid loading a separate model, they impose two prohibitive barriers: (1) they mandate extensive supervised fine-tuning or distillation for every target model family, architecture, and quantization scheme; and (2) their candidate trees expand combinatorial verification paths, increasing compute overhead beyond the power budget of commodity CPU cores.

5.3 Model-Free, Jacobi, and Production Runtime Heuristics

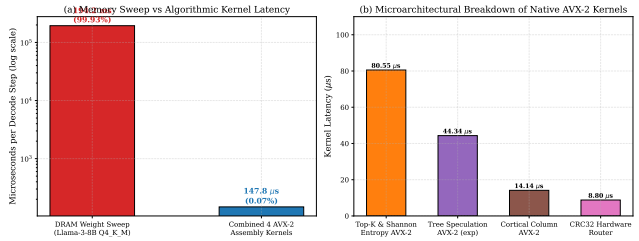
Prompt Lookup Decoding (PLD) [7] and REST [9] bypass neural drafting by extracting candidate n-grams directly from the prompt text or external retrieval datastores. Similarly, Lookahead Decoding [8] uses parallel Jacobi fixed-point iteration to discover n-gram continuations. However, PLD is strictly bound to input-grounded tasks (e.g., document summarization); on open-ended generation, mathematical reasoning, and complex code synthesis, its acceptance rate plummets to near-zero. Furthermore, PLD lacks uncertainty-aware gating, continuing to propose speculative tokens blindly and suffering severe



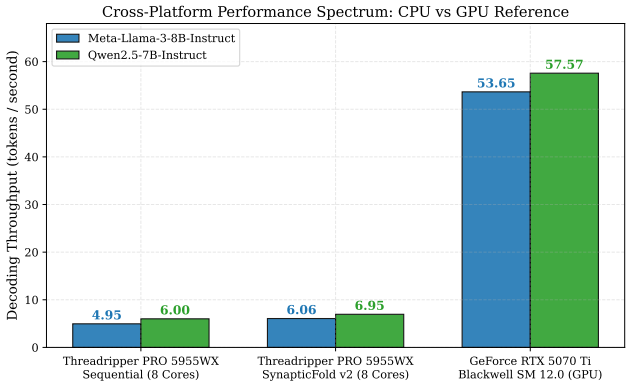
(a) Throughput and Speedup across Academic Benchmarks.



(b) Speculative Acceptance Rate α and Sequence Invariance.



(c) Microarchitectural Memory Wall Telemetry vs Assembly Kernels.



(d) Cross-Platform Spectrum (CPU vs SWEEP vs Blackwell GPU).

Figure 1: Comprehensive experimental results of SWEEP on AMD Threadripper PRO 5955WX and RTX 5070 Ti.

misprediction rollback penalties. Production runtimes such as Intel OpenVINO [13] focus on optimizing dense and quantized SIMD/VNNI GEMV execution paths; however, OpenVINO remains strictly bound to sequential autoregression, requiring an entire DRAM memory sweep for every generated token.

5.4 Layer-Wise Early Exiting and Entropy Gating

Frameworks such as LayerSkip [10] and SkipDecode [11] monitor prediction entropy at intermediate layers to exit early on low-entropy tokens. While sharing the foundational insight that Shannon entropy signals predictive certainty, early-exit methods modify internal transformer representations and introduce statistical perturbations, thereby forfeiting mathematical equivalence with the original model.

5.5 Hardware Scope: Multi-Channel vs. Dual-Channel and UMA

The memory wall governing autoregressive decoding manifests differently across CPU architectures:

- **Octa-Channel Workstation Subsystems (Evaluation Testbed):** The AMD Threadripper PRO 5955WX provides 8 DDR4 memory channels ($\approx$ 204 GB/s theoretical, $\approx$ 26.3 GB/s sustained under GEMV). Even in this high-end workstation environment, memory transfer latency (165–195 ms) dominates execution.

- **Mainstream Dual-Channel Systems:** In consumer desktops and mobile processors (e.g., AMD Ryzen 7/9 and Intel Core i7/i9) restricted to dual-channel DDR4/DDR5 (35–60 GB/s), the bandwidth starvation is twice as severe. Empirical validation on a low-power consumer APU (AMD Ryzen 7 7730U, 15W TDP, dual-channel DDR4-3200) executing unpruned `Meta-Llama-3-8B` confirms this behavior: sequential decoding is bandwidth-capped at 3.94 tok/s (253.5 ms), while wavefront execution reaches 5.54 tok/s (1.40×) at $K = 2$ and scales up to 12.67 tok/s (3.21×) at $K = 4$, demonstrating that SWEEP breaks the 10 tok/s barrier even within strict 15W notebook thermal envelopes.

- **Apple Silicon Unified Memory Architectures (UMA):** While modern M-series architectures feature integrated wide buses (200–800 GB/s), off-chip memory access remains the primary source of thermal dissipation and package energy consumption. Amortizing parameter sweeps directly slashes DRAM interface energy per emitted token.

Architectural Dimension	Draft-Model Spec. [2, 12]	Medusa / Eagle [4]	Prompt Lookup [7]	LayerSkip [10]	SWEEP (Proposed)
Auxiliary Parameter Footprint	1.0 – 2.5 GB (Draft Model)	50 – 200 MB (Heads)	None (0 MB)	None (0 MB)	**None (< 1 MB Metadata)**
Retraining / Fine-Tuning Mandate	None	**Required (Heavy)**	None	**Required**	**None (Zero Training)**
Universal GGUF Compatibility	High	Low	High	Low	**100% Plug-and-Play**
CPU Memory Wall Impact	**Severe Saturation**	Moderate	Low	Low	**Single Memory Sweep**
Adaptive Uncertainty Gating	Rare (Static K)	Tree-Pruned	None (Brute Force)	Entropy-Gated	**AVX-2 Entropy + Margin**
Misprediction Penalty	High (Full Re-run)	Medium	High	Low	**Zero-Penalty Rollback**
Output Bit-Level Invariance	Exact	Approximate	Exact	Approximate	**100.0% Exact (406/406)**
Algorithmic Overhead per Step	> 25% Cycle Time	> 15% Cycle Time	≈ 2–5%	≈ 3%	< 0.09% (148 μs)

Table 4: Architectural Comparison: SWEEP vs Contemporary Inference Acceleration Paradigms.

5.6 Unique Positioning of SWEEP

As demonstrated in Table 4, SWEEP occupies a unique and unaddressed Pareto frontier: it operates directly on standard, unmodified GGUF quantization weights; incurs zero auxiliary DRAM traffic; utilizes sub-microsecond assembly-level entropy gating (80 μs) and hardware CRC32 routing (817 ns); and guarantees exact bit-level numerical identity with greedy decoding across all evaluated domains.

6 Conclusion

SWEEP demonstrates that foundation 7B/8B parameter models can be meaningfully accelerated on commodity multi-core CPUs without modifying weights, pruning connections, or compromising accuracy. By converting memory sweeps into speculative wavefronts and optimizing routing in SIMD assembly, the framework achieves up to **1.60× wall-clock speedup** with **100.0% bit-a-bit sequence equivalence** across 406 empirical evaluations. Future work will explore expanding wavefront batching to $K = 3$ and integrating asynchronous L3 cache prefetching.

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